

Password Security Analysis Toolkit (Simulated Environment)

Abstract

This project demonstrates a simulated password security toolkit designed to analyze password strength and estimate attack feasibility in a controlled environment.

Introduction

Passwords are widely used for authentication but weak passwords can lead to serious security risks such as data breaches and unauthorized access.

Objectives

Develop a secure and ethical toolkit to simulate password attacks, analyze password strength, and recommend improvements.

Tools & Technologies

Python, hashlib, itertools, math library. All implementations are done in a simulated environment.

Methodology

The system includes dictionary generation, hash simulation, brute-force estimation, and entropy-based strength analysis.

System Flowchart

Input → Dictionary Generation → Hash Simulation → Brute Force Estimation → Strength Analysis → Report Generation

Sample Code Snippet

```
import hashlib
def hash_password(p):
    return hashlib.sha256(p.encode()).hexdigest()
```

Results & Analysis

Password	Strength	Entropy	Estimated Crack Time
admin123	Weak	28	< 1 minute
User2024	Medium	52	Few hours
Str0ng@Pass!	Strong	75	Years

Real-World Impact

Weak passwords can lead to credential stuffing, account takeover, and data breaches. Strong passwords and multi-factor authentication are essential.

Conclusion

This project highlights the importance of password strength and demonstrates how simulated analysis can improve security awareness.

Future Scope

Add GUI interface, integrate real-time password checks, and include MFA simulation.